



Intro Data Science

CHAPTERS

- Understand why data science is important
- Main step of data science
- Data analysis VS Machine learning

~2 contact hours

WHY THIS MODULE MATTERS

Provides a general introduction of data science and machine learning

What is Data Science?



“Data science is the application of **computational** and **statistical** techniques to address or gain insight into some problem in the **real world**”

[J. Zico Kolter, Carnegie Mellon University]

What is Data Science?



What is Data Science?



Image from
popsci.com

What is Data Science?

MODERN DATA SCIENTIST

Data Scientist, the sexiest job of the 21st century, requires a mixture of multidisciplinary skills ranging from an intersection of mathematics, statistics, computer science, communication and business. Finding a data scientist is hard. Finding people who understand who a data scientist is, is equally hard. So here is a little cheat sheet on who the modern data scientist really is.

MATH & STATISTICS

- ☆ Machine learning
- ☆ Statistical modeling
- ☆ Experiment design
- ☆ Bayesian inference
- ☆ Supervised learning: decision trees, random forests, logistic regression
- ☆ Unsupervised learning: clustering, dimensionality reduction
- ☆ Optimization: gradient descent and variants

DOMAIN KNOWLEDGE & SOFT SKILLS

- ☆ Passionate about the business
- ☆ Curious about data
- ☆ Influence without authority
- ☆ Hacker mindset
- ☆ Problem solver
- ☆ Strategic, proactive, creative, innovative and collaborative

PROGRAMMING & DATABASE

- ☆ Computer science fundamentals
- ☆ Scripting language e.g. Python
- ☆ Statistical computing packages, e.g., R
- ☆ Databases: SQL and NoSQL
- ☆ Relational algebra
- ☆ Parallel databases and parallel query processing
- ☆ MapReduce concepts
- ☆ Hadoop and Hive/Pig
- ☆ Custom reducers
- ☆ Experience with xaaS like AWS

COMMUNICATION & VISUALIZATION

- ☆ Able to engage with senior management
- ☆ Story telling skills
- ☆ Translate data-driven insights into decisions and actions
- ☆ Visual art design
- ☆ R packages like ggplot or lattice
- ☆ Knowledge of any of visualization tools e.g. Flare, D3.js, Tableau



Image from
marketingdistillery.com

What is Data Science?

Given **raw data** and a **problem statement**, choose a model to address the problem so that the **performance** on some evaluation **metric** is maximized.

Rigorous **testing** and **experimentation** has to be performed in order to **validate/refute hypotheses** and results.

Data Science Applications

Will stock X raise in the next few hours?



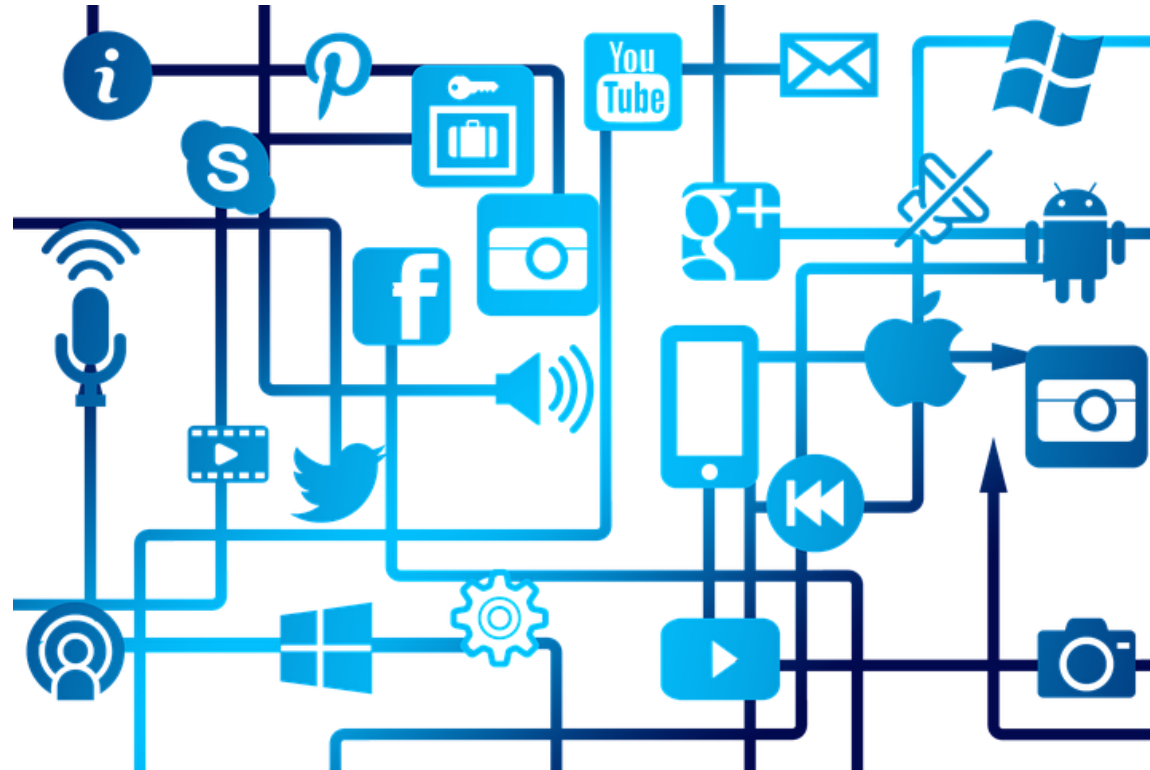
Data Science Applications

Should team T sign player X or Y?



Data Science Applications

Is customer X more similar to customer Y or customer Z?



Data Science Applications

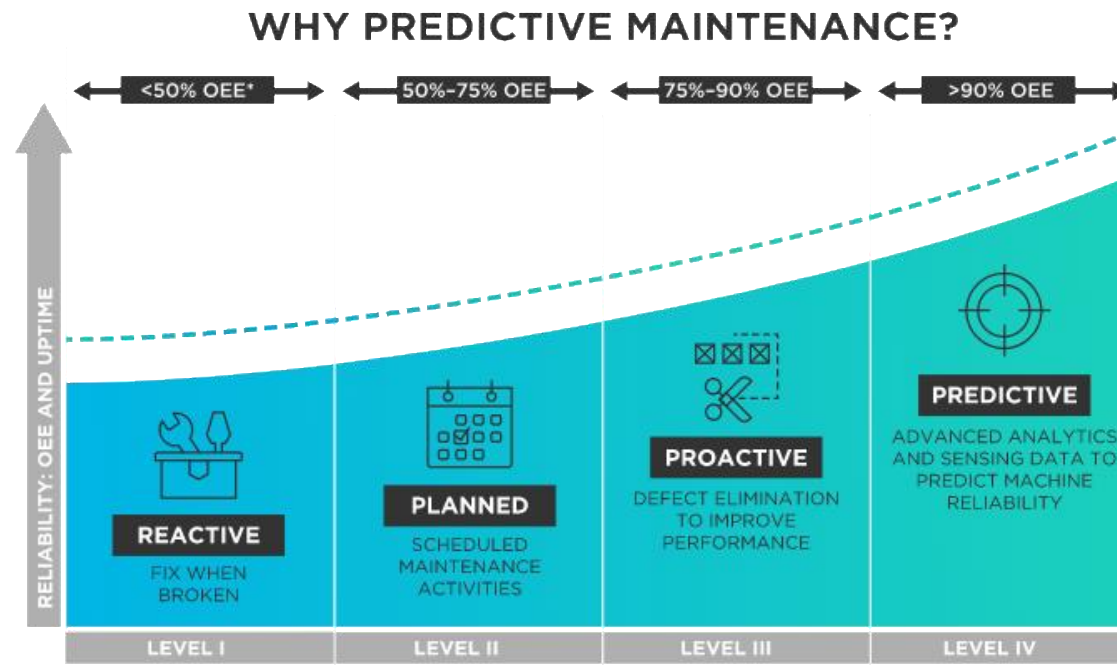
Should we grant a loan to X?

What are the main characteristics (features) driving the decision?



Data Science Applications

When/where should we perform equipment maintenance?



*OVERALL EQUIPMENT EFFECTIVENESS

[Imagine: Tibco]

Data Science Applications

Lead time prediction for supply chain optimization



Data Science Applications

Has procedure X improved company Y's production?



Data Science Applications

Does drug X cure disease Y?



Data Science Applications

Does gene X cause pathology Y?



Caveat



Observation, reason and experiment make up what we call the scientific method.

[Richard Feynman]

You do not get discoveries in the sciences by taking huge amounts of data, throwing them into a computer and doing statistical analysis of them... That's not the way you understand things... You have to have theoretical insights.

[Noam Chomsky, April 2014]

Data science requires **critical thinking**

Putting the science back in data science

Who Needs Data Science?



Jobs

Date posted

Remote

Research & Development

Higher Ed

AI/ML

Analytics

Easy Apply

Experience level

Learn more

Employment type

Company

Under 10 applicants

In my network

Verified

99+ results Italy How promoted jobs are ranked

**Risk Data Scientist** 

Credem Banca

Reggio nell'Emilia (Hybrid)

 9 connections work here

1 month ago

**Data Scientist** 

IPREL Progetti

Imola

 1 school alumni works here

2 weeks ago · Promoted

**Data Scientist** 

Onit

Forlì (Hybrid)

 1 connection works here

8 months ago

**Data Analytics Manager (in sede a Lido di Camaione)** 

Italia Gas E Luce

Tuscany, Italy (On-site)

 Actively reviewing applicants

2 weeks ago ·  Easy Apply · Promoted

**Data Scientist** 

Zucchetti

Turin (Hybrid)

 3 company alumni work here

1 week ago · Promoted

 Credem Banca

Risk Data Scientist 

Reggio nell'Emilia, Emilia-Romagna, Italy · 1 month ago · Over 100 people clicked apply

Responses managed off LinkedIn

Hybrid

Full-time

Apply

Save

Your profile and resume are missing some required qualifications



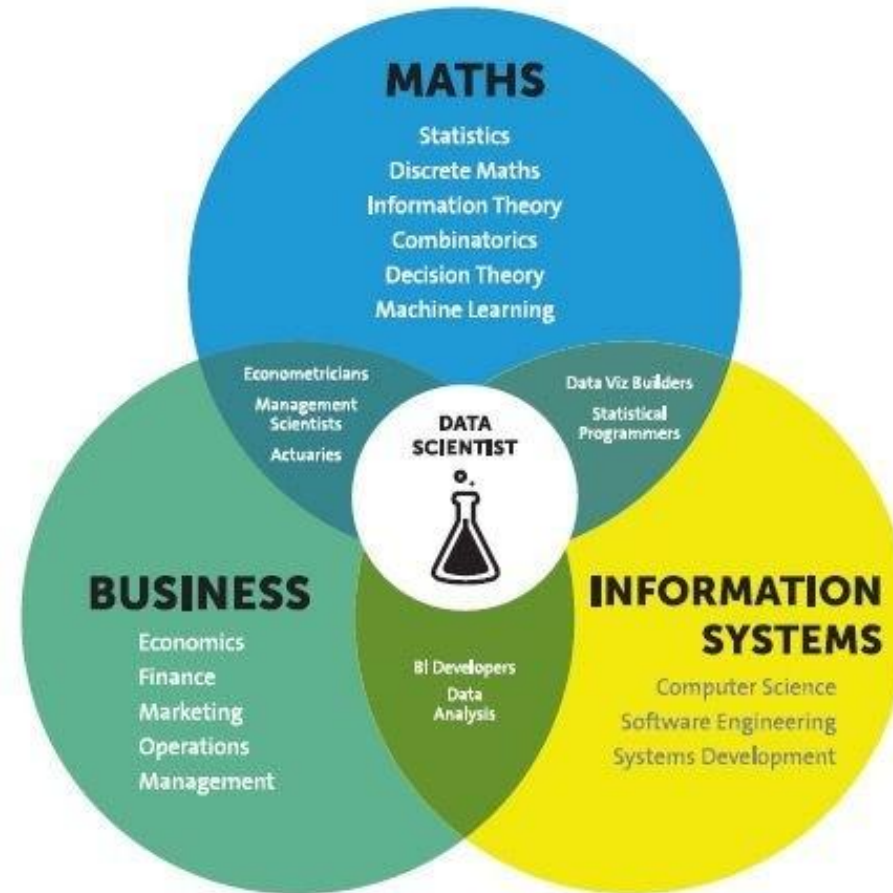
BETA · Is this information helpful?

See how you compare to over 100 others who clicked apply

Access exclusive applicant insights, see jobs where you have the highest chance of hearing back, and more.

 **Activate Premium for €0**

Related Fields



Business analytics requires a combination of expertise across business, information systems and mathematics.

Data Science Pipeline



Data science mainly needs three steps:

1. Data **collection** (pre-processing)
2. Data **analysis** (exploration, modeling, testing)
3. Data **presentation** (communication of results)

Data Collection



The first step in almost any data science application is to **collect some data**

Data could be available in several different ways:

- Plain text, Images, Audio, CSV, JSON, ...
- Database (SQL, ...)
- Web (HTTP requests, API, ...)

Data Collection



More on advanced data formats...

- How to deal with **high-dimensional** data?
- How to deal with **structured data** (e.g., graphs)?
- How to deal with **continuous data streams**?

Data Analysis



Exploration

- Take a first look at the data

Modeling

- Choose a model to be used

Testing

- Perform experiments to test model and hypothesis

Data Presentation



Present results of analysis in an **appropriate** way

Again, make use of data **visualization**

Give the **simplest** possible representation that conveys the message “Maximize information, minimize ink”

Data Presentation

“The ability to **take data** – to be able to **understand** it, to **process** it, to **extract value** from it, to **visualize** it, to **communicate** it's going to be a hugely important skill in the next decades, not only at the professional level but even at the educational level for elementary school kids, for high school kids, for college kids. Because now we really do have essentially free and ubiquitous data.”

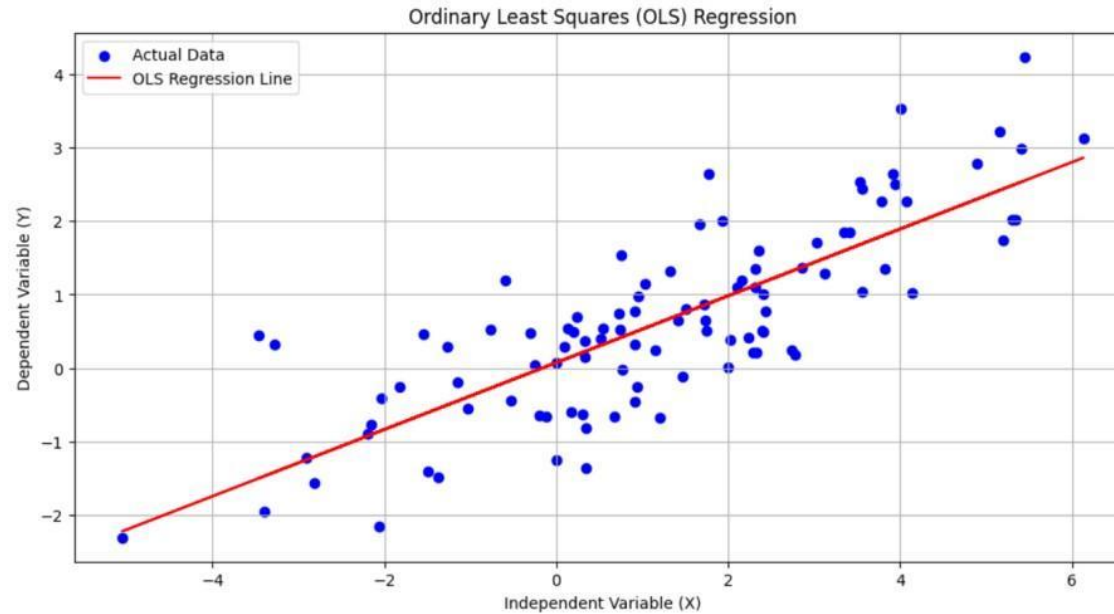
[Hal Varian, Google's Chief Economist]

Two Main Kinds of Analyses



- **Explanatory (or inferential) modeling:** the use of statistical models to estimate, interpret, and test relationships among variables, quantify uncertainty, and draw conclusions about an underlying population or theoretical mechanism.
 - *Statistics*
- **Predictive modeling:** the use of models to generate accurate predictions for new or unseen observations, with performance assessed primarily through out-of-sample validation and prediction error.
 - *Machine Learning*

Example: Linear Regression (OLS)



Explanatory (or inferential) modeling

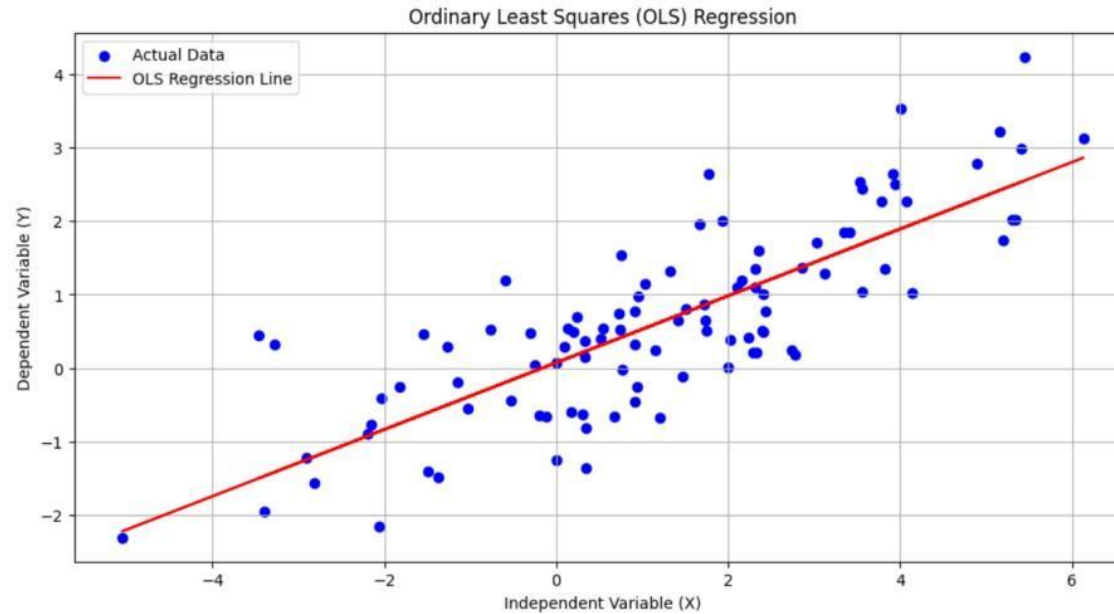
What is the relationship between X and Y?

How much variability of Y is explained by X?

How precisely can the relationship be estimated?

Is the observed relationship significant or it is by chance?

Example: Linear Regression (OLS)



Predictive modeling

How accurately can Y be predicted from X ?

How well does the model perform on new, **unseen** observations?

What is the expected **prediction** error?

Does the model **generalize beyond the data used to train it**?

Which variables contribute most to predictive performance?

Exercise!



00 - SuperStoreUS-2015.xlsx

- **Order ID** - A unique identifier for each order.
- **Customer ID** - A unique identifier for each customer.
- **Order Date** - The date of the order placement.
- **Ship Date** - The date the order was shipped.
- **Ship Mode** - The shipping mode for the order (e.g. standard, same-day).
- **Segment** - The customer segment (e.g. Consumer, Corporate, Home Office).
- **Region** - The region where the customer is located (e.g. West, Central, East).
- **Category** - The category of the product purchased (e.g. Furniture, Technology, Office Supplies).
- **Sub-Category** - The sub-category of the product purchased (e.g. Chairs, Desktops, Paper).
- **Product Name** - The name of the product purchased.
- **Sales** - The sales revenue for the product purchased.
- **Quantity** - The number of units of the product purchased.
- **Discount** - The discount applied to the product purchased.
- **Profit** - The profit generated by the product purchased.

- **Questions:**

- What is the total revenue generated by the store?
- Which category of products contributes the most to sales?
- How has the sales trend been for the past year?
- Which region has the highest sales and which one has the lowest?
- What is the average profit margin of the store?

For the next time...

Python3
IDE (VSCode)
Copilot / Codex / ...

Script (.py)
Documenti interattivi (.ipynb)

